WORKED SOLUTION

STEP II 1991, Question 15

Probability and Statistics • Linearity of Expectation

STEP II 1991, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q15

The key idea

Think of the pair (n_1, n_2) as a cell in an $m \times m$ grid, all m^2 cells equally likely. The cells with $|n_1 - n_2| = k$ lie on two diagonals parallel to the main one, so they are easy to count. After that, everything follows from *linearity of expectation*: a sum of differences has expected value equal to the sum of the expected values, even though neighbouring terms share a variable and are not independent. The invigilator's walk is the same kind of sum, except that the first leg starts from a fixed desk rather than a random one.

Solution

The distribution of $|n_1 - n_2|$

Since n_1 and n_2 are independent and each is uniform on $\{1, \ldots, m\}$, each of the m^2 ordered pairs (n_1, n_2) has probability $1/m^2$.

Fix k with $1 \le k \le m-1$. The pairs with $n_1 - n_2 = k$ are $(k+1, 1), (k+2, 2), \ldots, (m, m-k)$, which is $m - k$ pairs. By symmetry there are another $m - k$ pairs with $n_2 - n_1 = k$, and the two sets don't overlap because $k \neq 0$. Hence

$$\mathrm{P}(|n_1 - n_2| = k) = \frac{2(m-k)}{m^2},$$

as required. (For $k = 0$ there are m pairs, so $\mathrm{P}(|n_1 - n_2| = 0) = 1/m$. As a check, $\frac{1}{m} + \sum_{k=1}^{m-1} \frac{2(m-k)}{m^2} = \frac{1}{m} + \frac{m(m-1)}{m^2} = 1$.)

The expectation of $|n_1 - n_2|$

The value $k = 0$ adds nothing to the expectation, so

$$\begin{aligned}\mathrm{E}|n_1 - n_2| &= \sum_{k=1}^{m-1} k \cdot \frac{2(m-k)}{m^2} = \frac{2}{m^2}\left(m\sum_{k=1}^{m-1} k - \sum_{k=1}^{m-1} k^2\right) \\ &= \frac{2}{m^2}\left(\frac{m \cdot m(m-1)}{2} - \frac{(m-1)m(2m-1)}{6}\right) \\ &= \frac{2}{m^2} \cdot \frac{m(m-1)(3m-(2m-1))}{6} = \frac{(m-1)(m+1)}{3m} = \frac{m^2-1}{3m},\end{aligned}$$

using the standard sums $\sum_{k=1}^{n} k = \frac{1}{2}n(n+1)$ and $\sum_{k=1}^{n} k^2 = \frac{1}{6}n(n+1)(2n+1)$ with $n = m - 1$.

The case $m = 2$ with three integers

When $m = 2$ there are $2^3 = 8$ equally likely triples (n_1, n_2, n_3). Writing $S = |n_1 - n_2| + |n_2 - n_3|$:

(n_1, n_2, n_3)	111	112	121	122	211	212	221	222
S	0	1	2	1	1	2	1	0

The total is 8, so $\mathrm{E}(S) = 8/8 = 1$. The claimed value is $2(m^2-1)/(3m) = 2 \cdot 3/6 = 1$, which agrees.

Ask yourself The terms $|n_1 - n_2|$ and $|n_2 - n_3|$ both involve n_2. Can we still just add their expectations?

Yes. Linearity of expectation, $\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y)$, needs no independence at all. Each pair (n_i, n_{i+1}) on its own consists of two independent uniform integers, so each term has expectation $(m^2-1)/(3m)$. There are $r - 1$ terms, so

$$\mathrm{E}(|n_1 - n_2| + |n_2 - n_3| + \cdots + |n_{r-1} - n_r|) = \frac{(r-1)(m^2-1)}{3m}.$$

The case $r = 3$ is the result we just checked for $m = 2$.

The invigilator

Number the desks $1, 2, \ldots, m$ along the line, so desks i and j are a distance $d|i - j|$ apart. Let c_j be the number of the desk making the jth call, for $j = 1, \ldots, N + 1$. We model the callers as independent, each uniform on $\{1, \ldots, m\}$. This is how we read "equally likely to be called by any of the m candidates". Because candidates never call at the same time or during another call, the invigilator deals with the calls one after another and is standing at desk c_j when call $j + 1$ comes in. (If the same candidate calls twice running, that leg has length 0, and the formula allows for this.)

The total distance walked is therefore

$$D = d\,|c_1 - 1| + d\sum_{j=1}^{N} |c_{j+1} - c_j|.$$

The first leg. The invigilator starts at desk 1, and c_1 is uniform, so

$$\mathrm{E}|c_1 - 1| = \frac{1}{m}\sum_{i=1}^{m}(i-1) = \frac{m-1}{2}.$$

Notice that this is larger than $(m^2-1)/(3m)$. Starting at the end of the line makes the first walk longer than average.

The other N legs. Here $c_1, \ldots, c_{N+1}$ play the part of $n_1, \ldots, n_r$ with $r = N+1$, so the result above gives

$$\mathrm{E}\sum_{j=1}^{N} |c_{j+1} - c_j| = \frac{N(m^2-1)}{3m}.$$

Combining. By linearity again,

$$\begin{aligned}\mathrm{E}(D) &= d\left(\frac{m-1}{2} + \frac{N(m-1)(m+1)}{3m}\right)\\ &= \frac{d(m-1)}{6m}(3m + 2N(m+1)) = \frac{d(m-1)}{6m}[2N(m+1) + 3m],\end{aligned}$$

which is the required result.

Checks

- With $m = 1$ there is only one desk, so the invigilator never moves. The formula gives 0, as it should.
- With $N = 0$ there is only the first call, and the formula reduces to $d(m-1)\cdot 3m/(6m) = \frac{1}{2}d(m-1)$, the expected first leg.
- Exact enumeration of every possible sequence of callers confirms the formula for all $m \le 4$ and $N \le 3$. A simulation with $m = 10$, $N = 20$, $d = 1$ and 2×10^5 runs gave a mean of 70.50, against the formula's $\frac{9}{60}(440 + 30) = 70.5$.

Common slips

- Treating the first leg like the others. The invigilator starts at desk 1, not at a random desk, so that leg has expectation $\frac{1}{2}(m-1)$, not $(m^2-1)/(3m)$.
- Counting only $n_1 - n_2 = k$ and forgetting the factor 2 for $n_2 - n_1 = k$, or wrongly doubling the $k = 0$ case.
- Thinking that the terms of the sum must be independent before their expectations can be added.

Beyond STEP

The invigilator's position is a *Markov chain*: where they go next depends only on where they are now. Here the chain is so simple that each step "forgets" the past completely. In richer models (calls more likely from nearby desks, say) we would need the chain's stationary distribution to find the long-run average walk per call. The continuous analogue of the first result is that two independent uniform points on $[0,1]$ are on average $\frac{1}{3}$ apart, which is the limit of $(m^2-1)/(3m)$ divided by m. Grimmett and Welsh, *Probability: An Introduction*, covers both linearity of expectation and Markov chains.

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